

Correction des exercices

Chapitre 4

Exercice 1

```
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class Exercice1 {

    public static void main(String[] args) throws IOException {
        FileWriter fw = new FileWriter("calcul.txt");
        int n,d;
        StringBuilder ligne = new StringBuilder();
        do {
            Scanner s = new Scanner(System.in);
            System.out.print("Donner n: ");
            n = s.nextInt(); s.nextLine();
            if(n==0) {
                break;
            }
            d = n * n;
            ligne.append(n);
            ligne.append(" a pour carrée ");
            ligne.insert(ligne.length(), String.valueOf(d));
            ligne.append('\n');
            fw.write(ligne.toString());
            ligne.setLength(0);
        }while(true);
        fw.close();
        System.out.println("Lecture du fichier Texte :");
        FileReader fr = new FileReader("calcul.txt");
        int x;
        while((x = fr.read()) != -1) {
            System.out.print((char)x);
        }
        fr.close();
    }
}
```

Exercise 2 - solution 1

```

import java.io.*;
import java.util.Scanner;

public class Exercice2 {
    public static void main(String v[]) {
        final int longmax = 20;
        String nameFile = "infos.dat";
        String chnom, chprenom;
        int annee;
        int i,n=0;
        char[] nom = new char[longmax];
        char[] prenom = new char[longmax];
        for(i=0; i<longmax; i++) {
            nom[i] = ' ';
            prenom[i] = ' ';
        }
        try {
            FileOutputStream fos = new FileOutputStream(nameFile);
            DataOutputStream dos = new DataOutputStream(fos);
            Scanner s = new Scanner(System.in);
            while(true) {
                n++;
                System.out.println("Personne: " + n);
                System.out.print("Nom: ");
                chnom = s.nextLine();
                if(chnom.equals("")) {
                    break;
                }
                System.out.print("Prénom: ");
                chprenom = s.nextLine();
                System.out.print("Année de naissance: ");
                annee = s.nextInt(); s.nextLine();
                //Copier le nom vers un tableau de caractères
                for(i=0; i<longmax && i<chnom.length(); i++) {
                    nom[i] = chnom.charAt(i);
                }
                //Copier le prénom vers un tableau de caractères
                for(i=0; i<longmax && i<chprenom.length(); i++) {
                    prenom[i] = chprenom.charAt(i);
                }
                //écriture dans le fichier
                for(i=0; i<longmax; i++) { dos.writeChar(nom[i]); }
                for(i=0; i<longmax; i++) { dos.writeChar(prenom[i]); }
                dos.writeInt(annee);
            }
            dos.close();
            fos.close();
        } catch (FileNotFoundException e) {
            System.out.println(e.getMessage());
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}

```

```

//Lecture à partir du fichier infos.dat
try {
    FileInputStream fis = new FileInputStream(nameFile);
    DataInputStream dis = new DataInputStream(fis);
    while (dis.available() > 0 ) {
        for(i=0; i<longmax; i++) { nom[i] = dis.readChar(); }
        for(i=0; i<longmax; i++) { prenom[i] = dis.readChar();}
        annee = dis.readInt();
        System.out.println(String.valueOf(nom) +
            String.valueOf(prenom) +
            String.valueOf(annee));
    }
    dis.close();
    fis.close();
} catch (FileNotFoundException e) {
    System.out.println(e.getMessage());
} catch (IOException e) {
    System.out.println(e.getMessage());
}
}
}

```

Exercice 2 - solution 2

```

import java.io.*;
import java.util.Scanner;

public class Exercice2_UTF {
    public static void main(String v[]) {
        final int longmax = 20;
        String nameFile = "infos2.dat";
        String chnom, chprenom;
        int annee;
        StringBuilder sb = new StringBuilder();
        int i,n=0;
        char[] nom = new char[longmax];
        char[] prenom = new char[longmax];
        for(i=0; i<longmax; i++) {
            nom[i] = ' ';
            prenom[i] = ' ';
        }
        try {
            FileOutputStream fos = new FileOutputStream(nameFile);
            DataOutputStream dos = new DataOutputStream(fos);
            Scanner s = new Scanner(System.in);
            while(true) {
                n++;
                System.out.println("Personne: " + n);
                System.out.print("Nom: "); chnom = s.nextLine();
                if(chnom.equals("")) {
                    break;
                }
            }
        }
    }
}

```

```

    }
    System.out.print("Prénom: "); chprenom = s.nextLine();
    System.out.print("Date de naissance: ");
    annee = s.nextInt(); s.nextLine();

    sb.append(chnom);
    for(int k = chnom.length(); k < 20; k++) {
        sb.append('.');
    }
    sb.append(chprenom);
    for(int k = (20+chprenom.length()); k < 40; k++) {
        sb.append('.');
    }
    sb.append(String.valueOf(annee));
    System.out.println(sb.toString());
    dos.writeUTF(sb.toString());
    sb.setLength(0); //vider StringBuilder
}
System.out.println("Fin de saisie\n");
dos.close();
fos.close();
} catch (FileNotFoundException e) {
    System.out.println(e.getMessage());
} catch (IOException e) {
    System.out.println(e.getMessage());
}

//Lecture à partir du fichier infos2.dat
try {
    FileInputStream fis = new FileInputStream(nameFile);
    DataInputStream dis = new DataInputStream(fis);
    while(dis.available() > 0 ) {
        System.out.println(dis.readUTF());
    }
    dis.close();
    fis.close();
} catch (FileNotFoundException e) {
    e.printStackTrace();
} catch (IOException e) {
    e.printStackTrace();
}

}
}

```